

OCPP 1.6-J & 2.1

USER & OPERATOR MANUAL

RELEASE V2.4

OCPP Charge Point Management System

Enterprise EV Charging Network Operations, Ground Plans, Tariffs, Invoicing, RFID & Driver Companion Guide

DOCUMENT TARGET

Charge Point Operators (CPOs), Facility Managers & Drivers

SYSTEM VERSION

v2.4 Enterprise Edition

PUBLICATION DATE

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PUBLISHER

The Charge Grid

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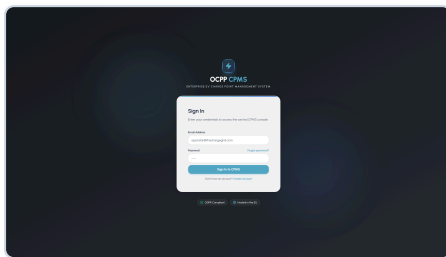
- 16. Mobile Driver Companion Web Application**

1. Getting Started & Authentication Security

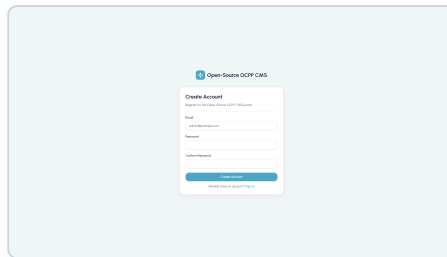
The **OCPP-CPMS** dashboard provides a secure web interface for operating electric vehicle charging infrastructure. Access the platform by navigating to your organization's URL (e.g. <https://ocpp.thechargegrid.com> or <http://localhost:3002>).

1.1 Login & Two-Factor Authentication (2FA TOTP)

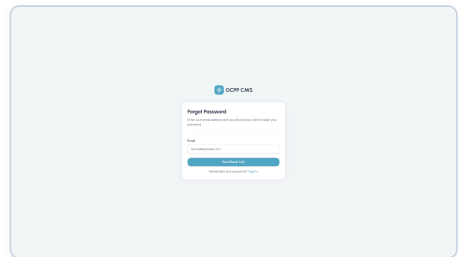
- **Standard Authentication:** Input your registered corporate email and password. Passwords are securely hashed with salted bcrypt.
- **Two-Factor Authentication:** If 2FA is active on your profile, you will be prompted for a 6-digit Time-Based One-Time Password (TOTP) from an authenticator app (Google Authenticator, Microsoft Authenticator, 1Password).
- **Password Reset & Email Verification:** Automated self-service workflows exist for forgotten credentials and initial account email validation.



Secure Operator Login



Driver / User Registration



Self-Service Password Reset

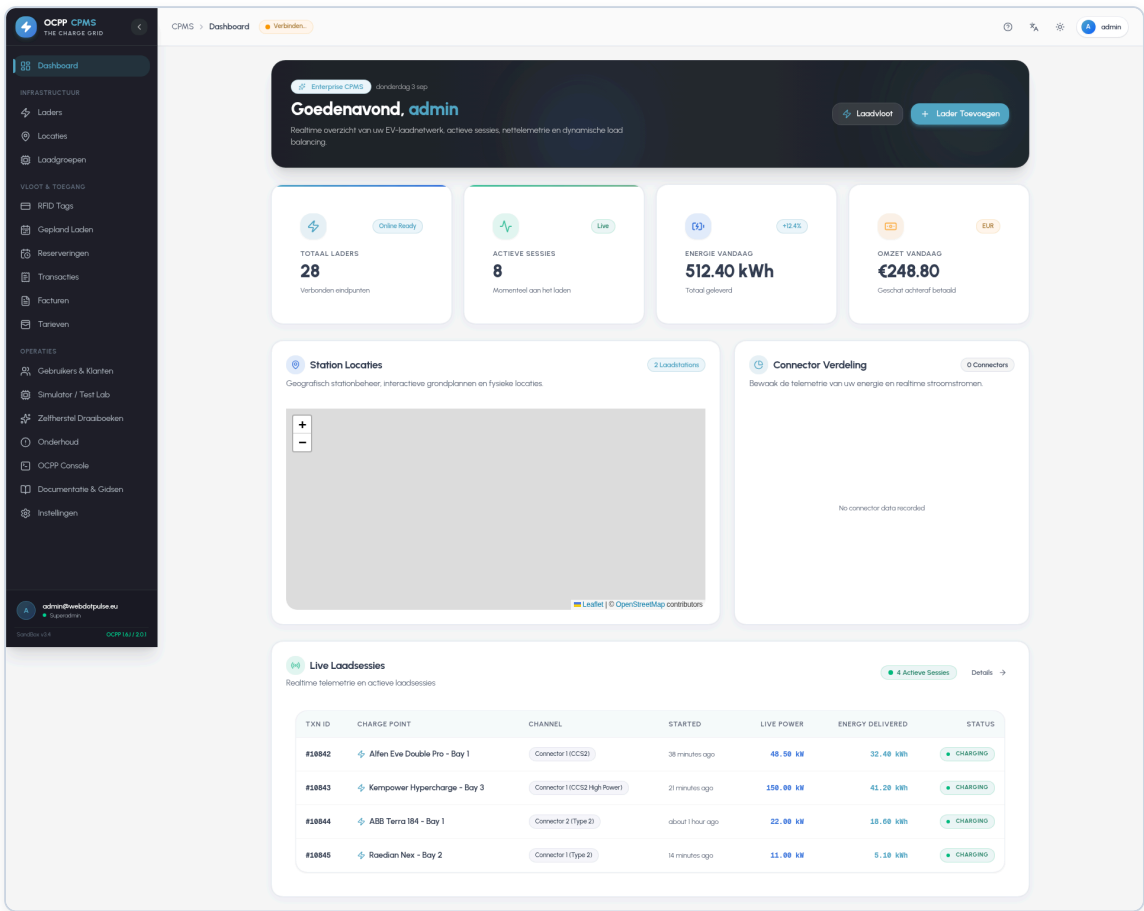
2. Executive Dashboard & Network KPIs

The **Executive Dashboard** (`/dashboard`) provides an instant operational summary across all connected charging stations:

LIVE TELEMETRY OVERVIEW

Metrics update automatically in real-time via Socket.IO WebSocket streams without requiring manual page refreshes.

- **Total Energy Delivered (kWh):** Real-time accumulation of kilowatt-hours dispensed across all active and completed sessions.
- **Active Charging Sessions:** Count of vehicles actively drawing or negotiating power.
- **Fleet Online Health:** Percentage and status breakdown of online vs offline charge points.
- **Geospatial Station Map:** Interactive Leaflet map displaying charging locations with status pins (Available, Charging, Faulted, Offline).
- **24-Hour Load Profiles:** Visual hourly charts detailing peak demand and energy utilization trends.



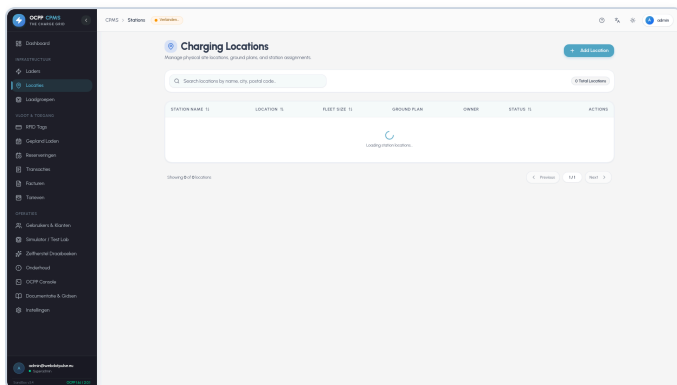
Executive Dashboard with Live Fleet Metrics and Map View

3. Charging Stations & Interactive 2D Ground Plans

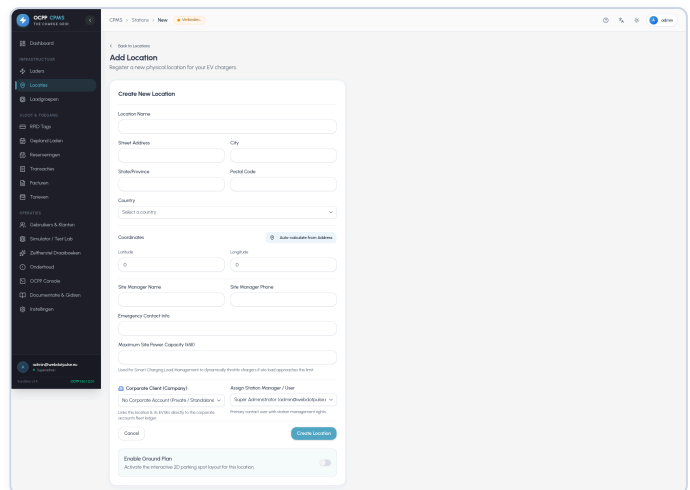
A **Station** represents a physical charging site containing one or more physical charge points with specific geographical, electrical, and operational parameters.

3.1 Station Directory & Management (/stations)

- Create and edit stations with address, GPS latitude/longitude, total power capacity caps, and emergency contacts.
- Filter stations by active status, city, or assigned charge group cluster.



Stations Directory & Geospatial Map

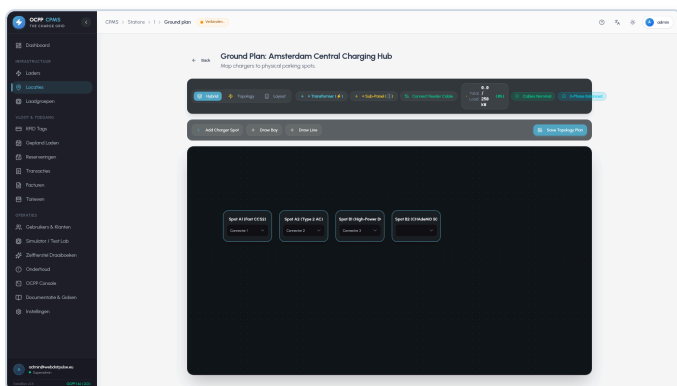


Create Station Modal Form

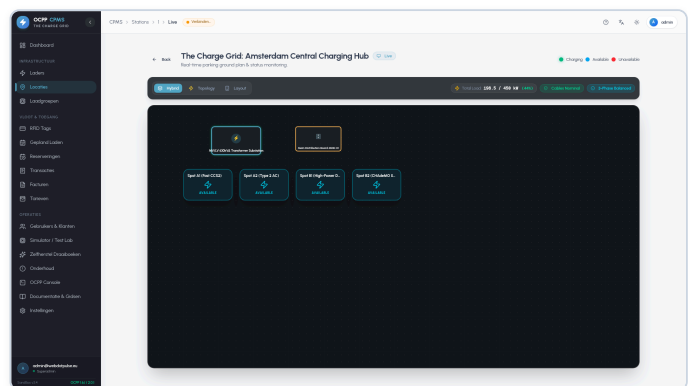
3.2 2D Ground Plan Canvas & Live Floor Monitor

Enable Ground Plan on any station to build high-precision visual parking bay layouts:

- **2D Canvas Editor (/stations/[id]/ground-plan)**: Drag and drop parking bays, rotate bays at 45° angles, draw pedestrian paths and canopy shelters, map physical charger sockets to spots, and draw electrical feeder topology lines.
- **Live Floor Monitor (/stations/[id]/live)**: Real-time floor plan showing live spot occupancy, charging wattage, delivered kWh, 3-phase balance (L1/L2/L3), and driver RFID tag badges.



Interactive 2D Ground Plan Canvas Builder

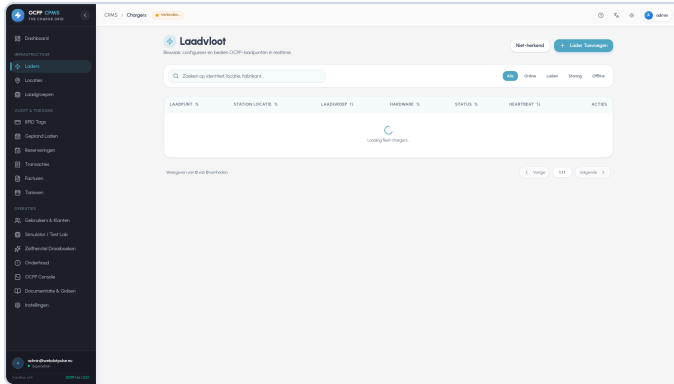


Live Station Floor Plan Monitor with Telemetry

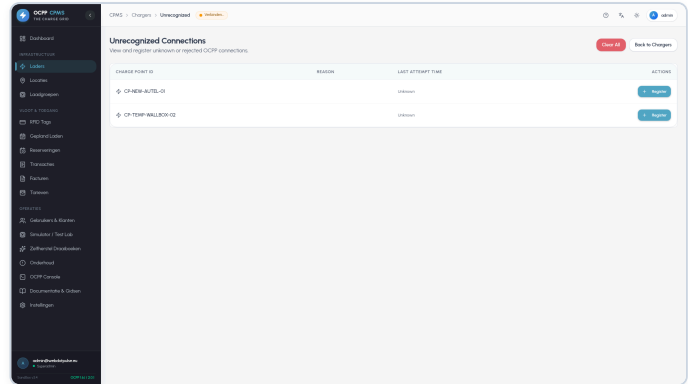
4. Chargers Fleet, EVSE Connectors & Remote Controls

4.1 Fleet Directory (/chargers)

Track all physical charging hardware connected to the central system with OCPP protocol version, firmware build, serial number, and live state.



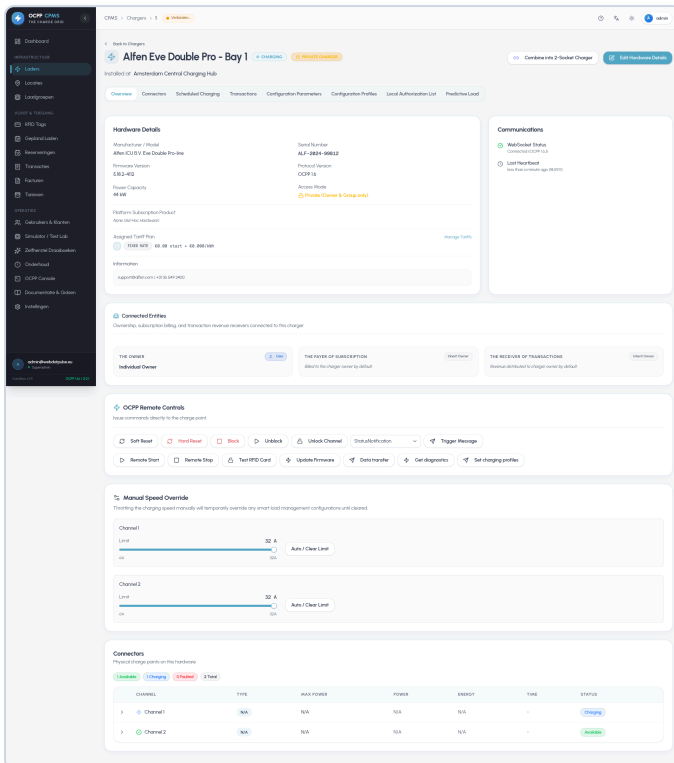
Chargers Fleet Directory



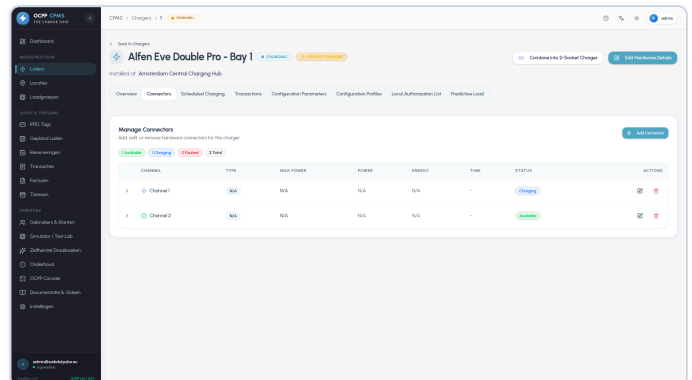
Unrecognized Chargers Auto-Discovery Queue

4.2 Charger Details & Remote Control RPCs (/chargers/[id])

- **Remote Control Commands:** Remote Start Transaction, Remote Stop Transaction, Soft/Hard Reset, Unlock Connector, and Change Availability (Operative/Inoperative).
- **Connectors Tab:** Configure plug types (Type 2, CCS2, CHAdeMO), phase types (1-phase / 3-phase / DC), and max voltage/ampere ratings.
- **Predictive Load Tab:** Inspect 24-hour solar irradiance forecasts and dynamic power limit schedules.



Charger Detail & Remote Control Console



EVSE Connectors Configuration Tab

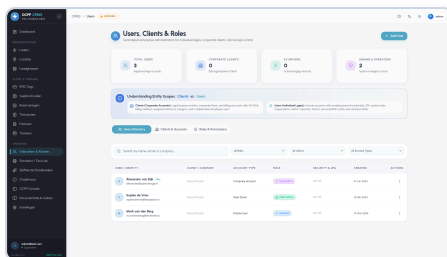
5. User Accounts, Corporate Clients & Granular RBAC

5.1 Corporate B2B Clients vs Individual Users

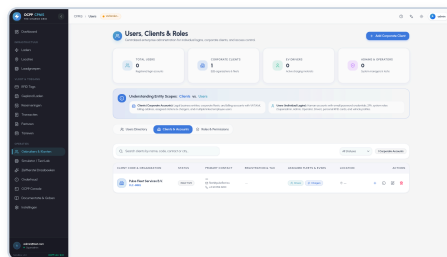
- Corporate Clients:** Business entities holding legal company names, VAT/KvK registrations, billing addresses, SEPA mandates, assigned charging hubs, and employee fleet drivers.
- Individual Users:** Driver accounts with login credentials, 2FA settings, assigned RFID cards, and vehicle battery profiles.

5.2 Roles & Permissions Matrix

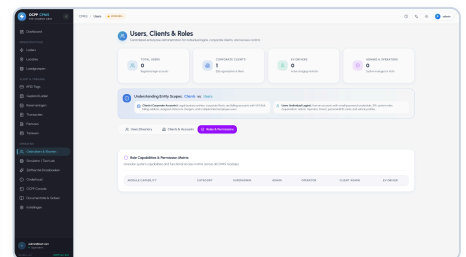
The platform enforces a 5-tier role hierarchy (Superadmin, Platform Admin, Operator/Technician, Client Admin, User/Driver) across all operational modules.



Users Accounts Directory



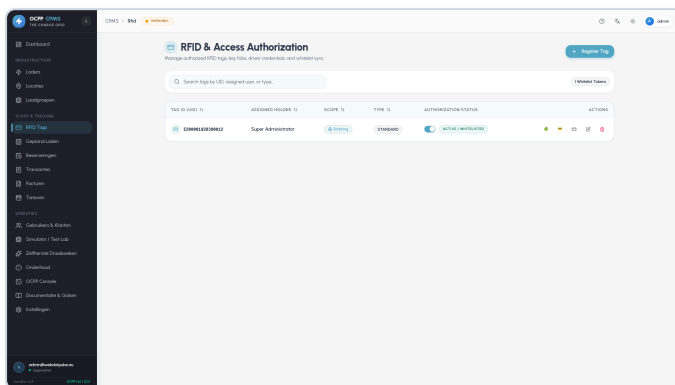
Corporate B2B Clients Hub



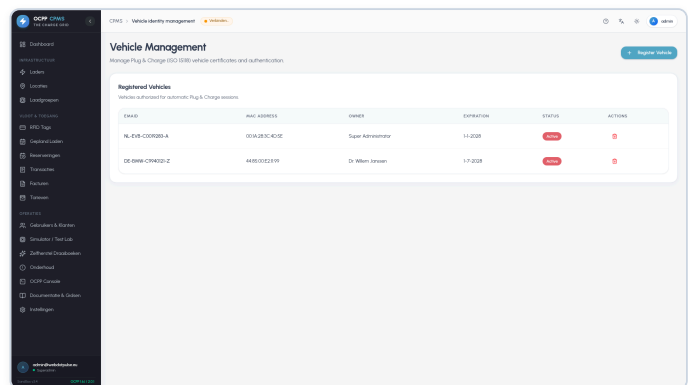
Roles & Capabilities Matrix

5.3 RFID Whitelist & ISO 15118 Plug & Charge

Manage RFID cards by `idTag` with instant hardware cache synchronization, and manage ISO 15118 contract certificates (EMAID) for automated Plug & Charge authorization.



RFID Whitelist Directory



ISO 15118 Plug & Charge Contract Certificates

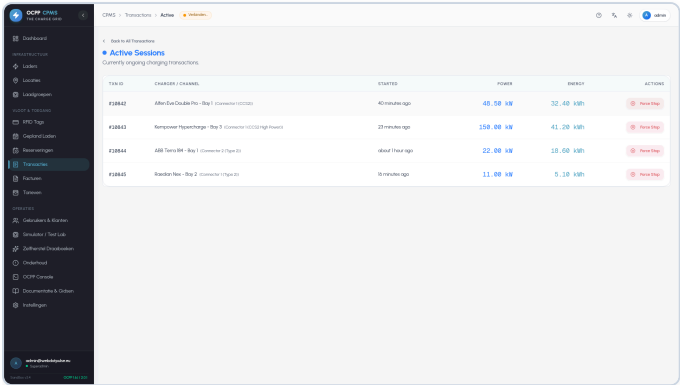
6. Active Sessions & Transaction History

6.1 Live Active Sessions (/transactions/active)

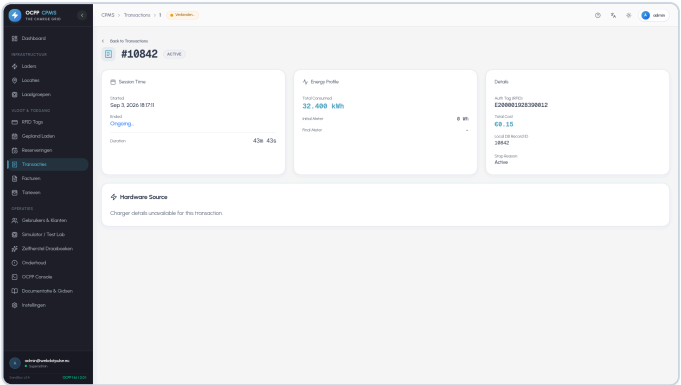
Monitor ongoing charging sessions in real-time with continuous duration timers, live wattage draw (kW), consumed energy (kWh), and dynamic cost accumulation.

6.2 Historical Records & Itemized Receipts (/transactions)

Search completed transaction records by date range, station, charger, card tag, or driver account, and inspect itemized VAT receipts.



Live Active Sessions Monitor

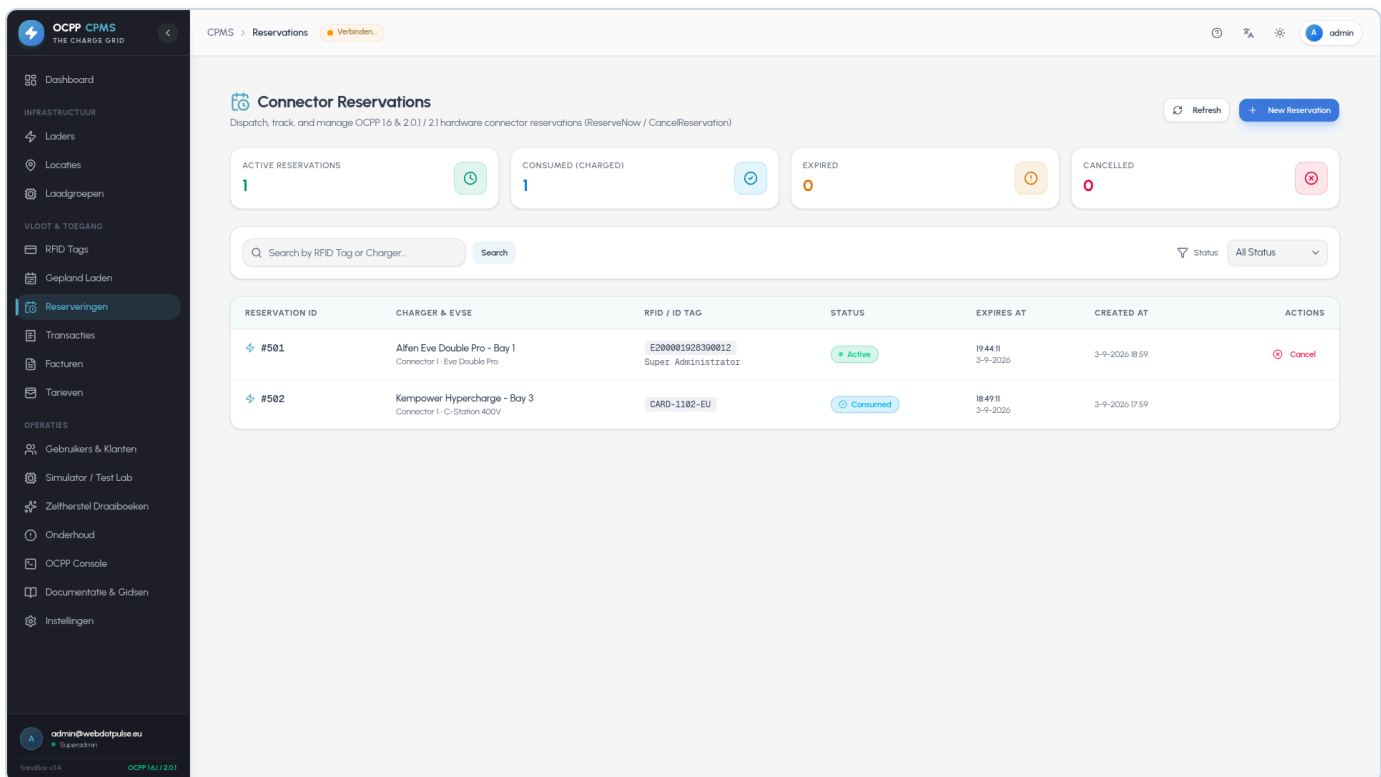


Detailed Transaction Tax Receipt

7. Reservations Manager

The **Reservations Manager** (/reservations) allows operators and drivers to schedule exclusive charging windows:

- Reserve specific EVSE sockets for specified arrival times.
- Transmits OCPP **ReserveNow** commands to place physical hardware in **Reserved** state.
- Configurable reservation expiry windows and automatic cancellation on no-show.



OCPP CPMS
THE CHARGE GRID

CPMS > Reservations Verbinden...

Connector Reservations
Dispatch, track, and manage OCPP 1.6 & 2.0.1 / 2.1 hardware connector reservations (ReserveNow / CancelReservation)

Refresh + New Reservation

ACTIVE RESERVATIONS 1 CONSUMED (CHARGED) 1 EXPIRED 0 CANCELLED 0

Search by RFID Tag or Charger... Search Status: All Status

RESERVATION ID	CHARGER & EVSE	RFID / ID TAG	STATUS	EXPIRES AT	CREATED AT	ACTIONS
#501	Alfen Eve Double Pro - Bay 1 Connector 1 - Eve Double Pro	E200001928390012 Super Administrator	Active	19:44:11 3-9-2020	3-9-2020 18:59	Cancel
#502	Kempower Hypercharge - Bay 3 Connector 1 - C-Station 400V	CARD-1102-EU	Consumed	18:49:11 3-9-2020	3-9-2020 17:59	

admin@webdatapulse.eu
Superadmin

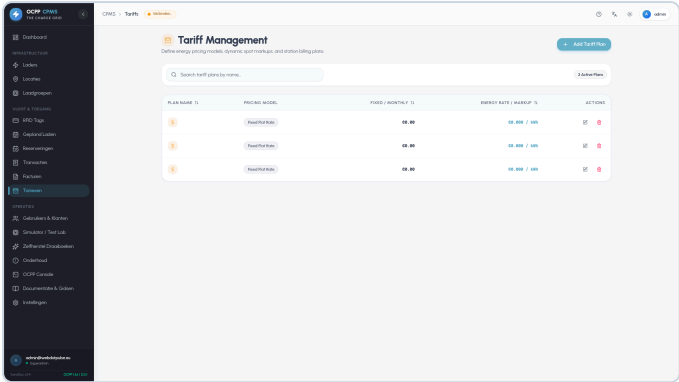
Sandbox v0.4 OCPP 1.6J / 2.0.1

Reservations Management Console

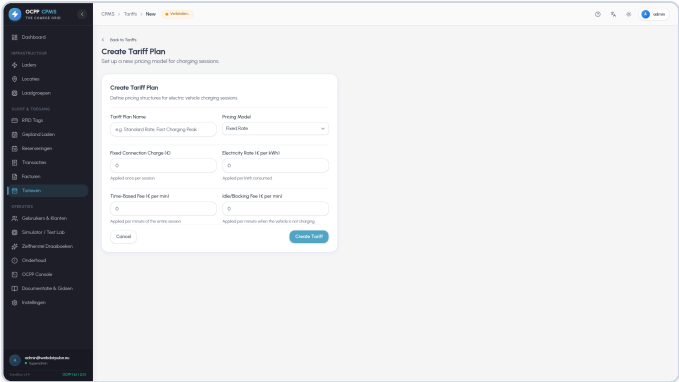
8. Tariffs & Dynamic EPEX Spot Pricing

8.1 4-Part Tariff Pricing Matrices (/tariffs)

FEE COMPONENT	UNIT	DESCRIPTION
Connection Fee	€ / session	Fixed one-time starting charge when session begins.
Energy Fee	€ / kWh	Price per kWh consumed (flat rate or dynamic spot).
Time Fee	€ / hour	Duration charge applied while vehicle is actively plugged in.
Idle Fee	€ / hour	Penalty surcharge after battery reaches 100% SoC.



Tariffs Pricing Structures

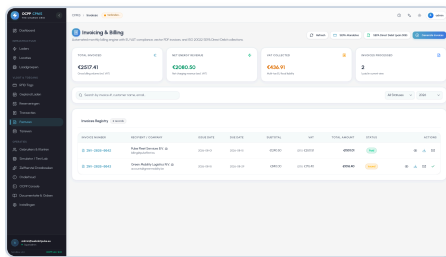


Create Tariff Modal Form

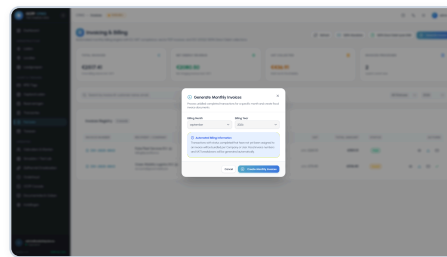
9. Enterprise Invoicing ("Facturen") & ISO 20022 SEPA

The **Invoices Suite** (/invoices) provides automated end-of-month billing runs for corporate accounts and private subscribers:

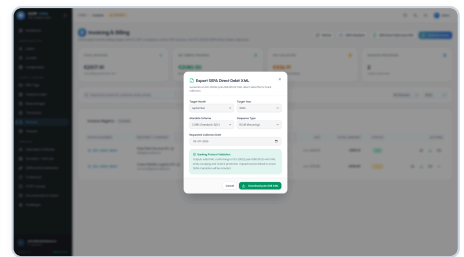
- **Billing Ledger:** Aggregates unbilled charging sessions and displays turnover, subtotal, and 21% VAT metrics.
- **SEPA Direct Debit XML (pain.008.001.02):** Generate standardized ISO 20022 banking direct debit files for direct upload to banking portals.
- **SEPA Mandate Management:** Track B2B and CORE Direct Debit mandates with Unique Mandate References (UMR).



Invoicing Ledger & Turnover KPIs



Monthly Batch Invoicing Wizard



ISO 20022 SEPA Direct Debit Export

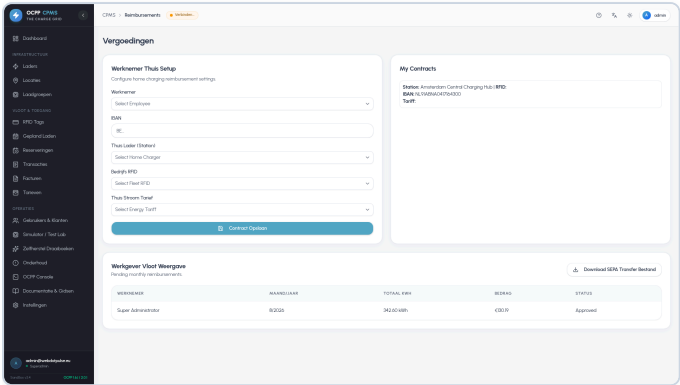
10. Home Reimbursements & Public Payments

10.1 Employee Home Reimbursements (/reimbursements)

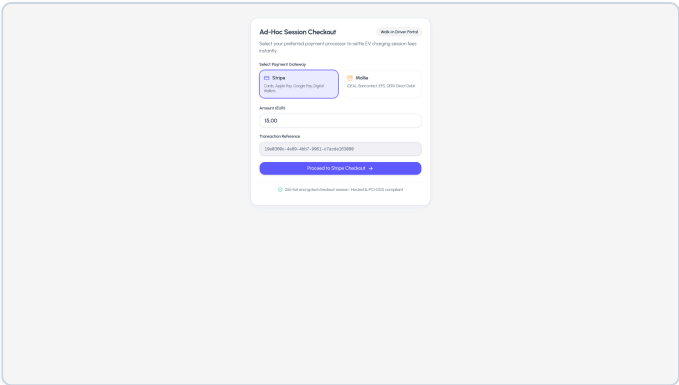
Calculates employee home charging kWh and generates **ISO 20022 SEPA Credit Transfer XML** (pain.001.001.03) for bulk corporate reimbursement payouts.

10.2 Public Walk-In Payments (Stripe & Mollie)

Ad-hoc walk-in drivers scan QR codes at charging bays to pay via Stripe (Credit Card, Apple Pay, Google Pay) or Mollie (iDEAL, Bancontact, EPS).



Employee Home Reimbursements & SEPA



Public Ad-Hoc Checkout Portal

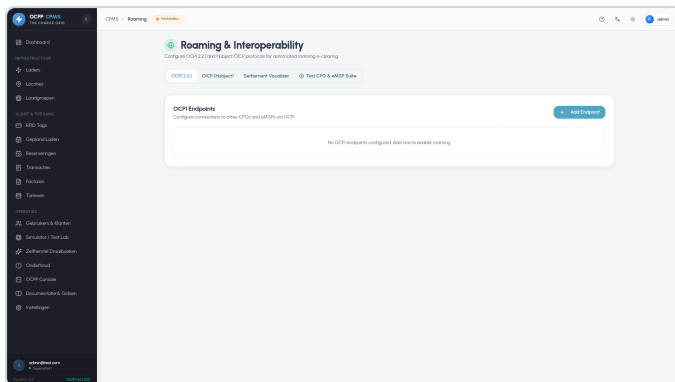
11. Roaming Hubs & V2G Smart Grid

11.1 OCPI 2.2.1 & Hubject OICP Roaming (/roaming)

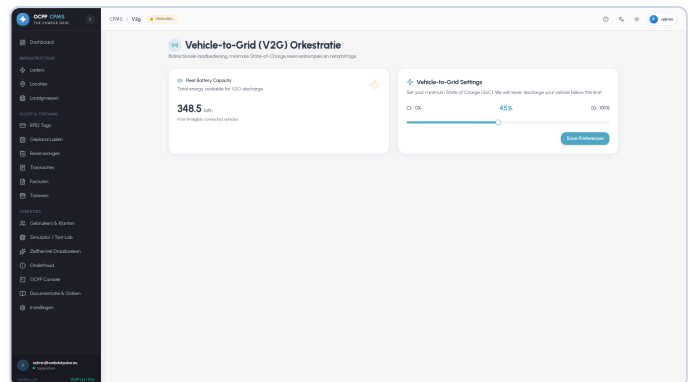
Synchronize locations, tariffs, sessions, tokens, and CDRs with international roaming clearinghouses.

11.2 V2G Battery Orchestration (/v2g)

Manage bidirectional power flow from EV batteries back to local buildings during high-tariff grid hours while maintaining minimum driver SoC reserves.



OCPI 2.2.1 Roaming Hubs

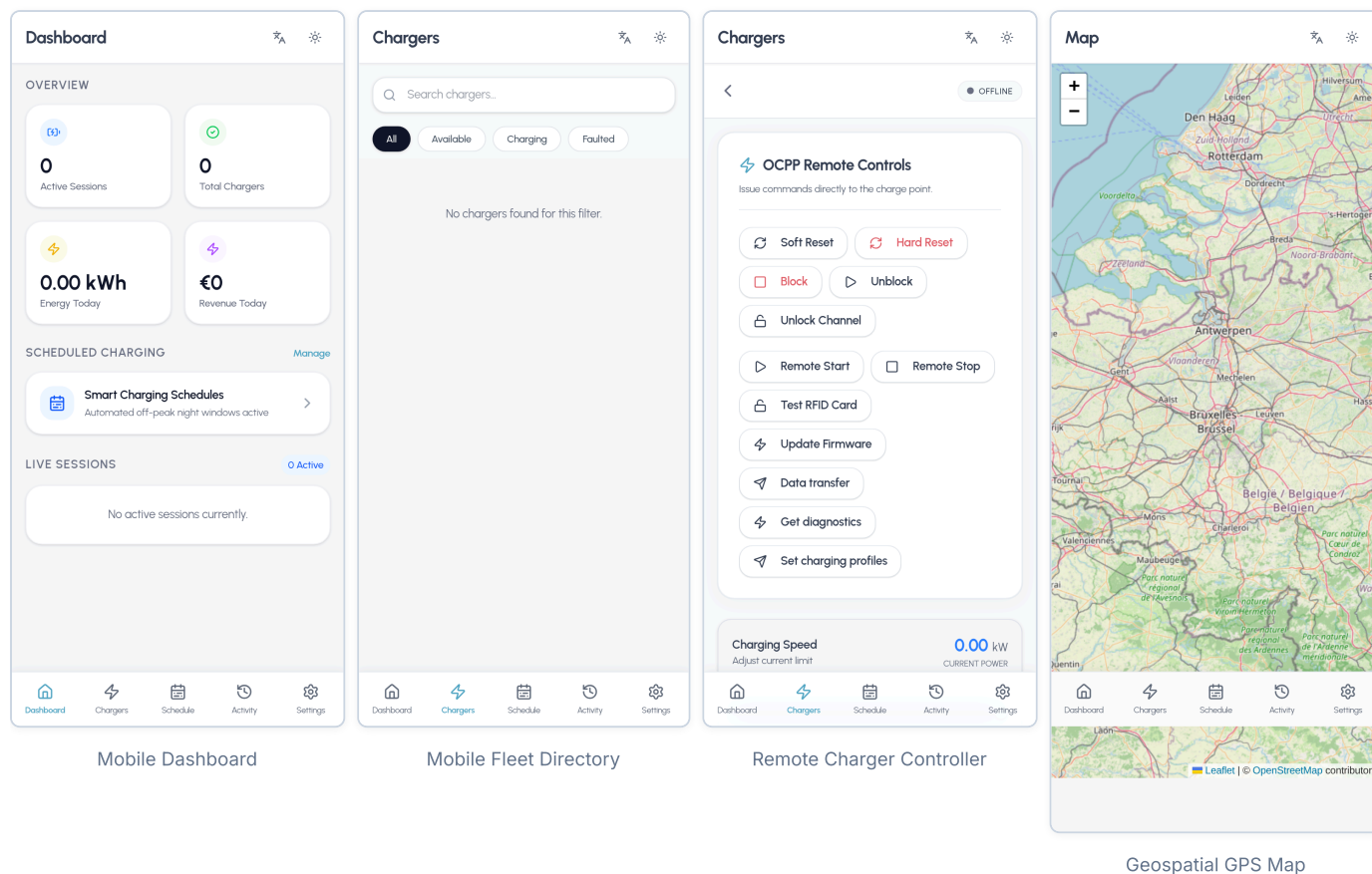


V2G Fleet Battery Orchestration

12. Mobile Driver Companion Web Application

The dedicated mobile web app (/mobile) provides a responsive smartphone experience for drivers on the road:

- **Mobile Dashboard** (/mobile/dashboard): Active charging widget, nearest station finder, and recent receipt history.
- **Mobile Fleet** (/mobile/chargers): List of available chargers with connector plug types and distance.
- **Remote Controller** (/mobile/chargers/[id]): Start and stop charging sessions directly from your phone.
- **Geospatial Map** (/mobile/map): Interactive GPS routing and charger availability pins.



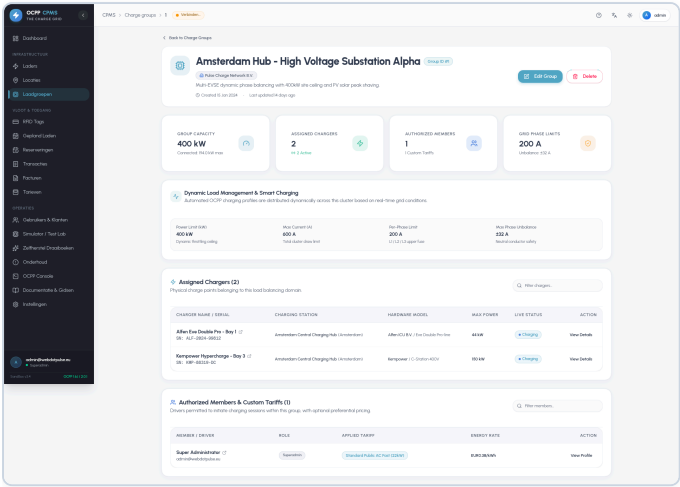
13. Dynamic Load Balancing & Auto-Heal Playbooks

13.1 Dynamic Load Balancing & Phase Allocator (/charge-groups)

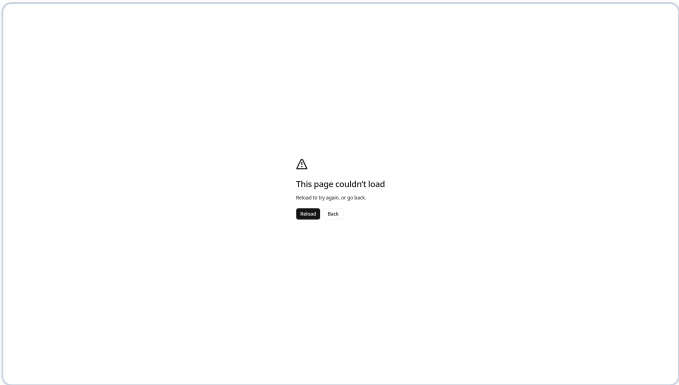
Distribute site electrical capacity, prevent transformer overload, and maintain phase balance across active chargers.

13.2 Automated Recovery Playbooks (/auto-heal-playbooks)

Automate multi-step remediation procedures to resolve connector lock failures, communication timeouts, and mechanical errors.



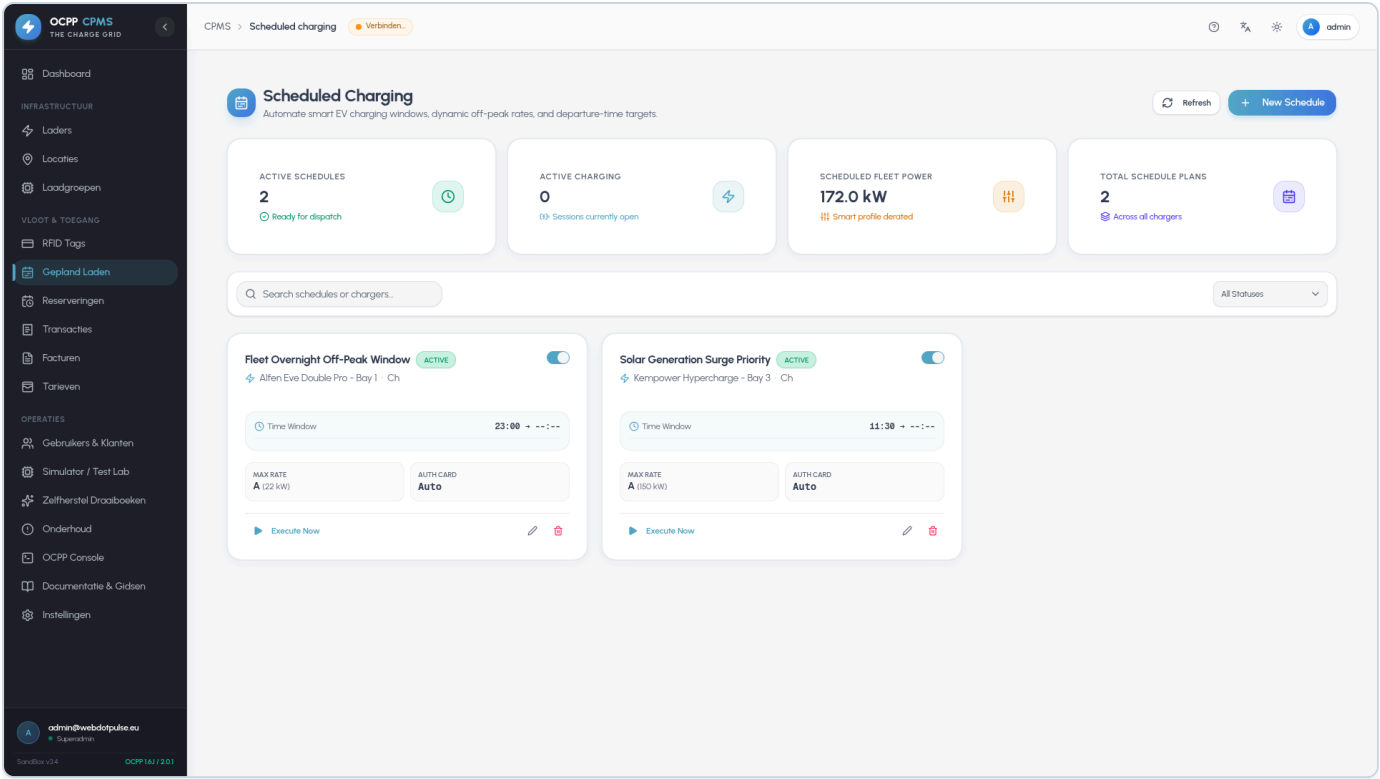
Charge Group Phase Allocation Detail



Automated Recovery Playbooks

13.3 Scheduled Smart Charging Calendar (/scheduled-charging)

Align fleet charging with off-peak spot rates, solar yield windows, and scheduled departure times.



Scheduled Smart Charging Calendar & Time-of-Use Matrix